A) **ANCHOR SCENES (3–6)**

SCENE #1: Curiosity in the Barley Field

Summary: A young Morten Meldal lies hidden in his grandfather's barley field on Bornholm, Denmark, observing a miniature world of insects and wildflowers under the crop canopy. He wonders how nature's diversity came to be, sparking his first scientific questions . This childhood scene illustrates the birth of his curiosity about the "mysteries of life and the universe" .

Date/Place: Circa 1960s, Grandparents' farm in Bornholm, Denmark.

People: Morten Meldal (as a child), with family nearby (grandfather's farm).

Concrete details: *"You have this roof of barley on top, and a microscopic world underneath"* with poppies, weeds, and big green grasshoppers. Young Morten is "thinking about, how does this evolve? Where does it come from?" . The farm's overseer routinely spanks his own kids "for what they could have done," which makes Morten grateful for the freedom to explore that his own family gives him . The blue sky above the barley and the teeming life below leave a vivid imprint on his memory.

Why it matters: This scene plants the seed of scientific curiosity and love of nature that will drive Meldal's career. It humanizes the origins of his passion for understanding biology and chemistry . A reader can see the future chemist as an observant boy fascinated by life's complexity.

Evidence quality: Primary (autobiographical) – Meldal himself recounts these memories in interviews and his Nobel biography . The details come directly from his own recollection, lending authenticity.

Myth risk: Low – Though a poetic memory, it's documented in his first-person accounts. No signs of embellishment beyond his own vivid storytelling.

Sources: Nobel interview 2022 ; Nobel autobiographical sketch .

SCENE #2: Inventing a Peptide Synthesizer in a Garage

Summary: In the late 1980s, Meldal takes a hands-on approach to a technical challenge by designing and building a multi-column peptide synthesizer himself. Lacking a commercial device to rapidly create many peptides, he uses a Teflon block and a drill in his neighbor's garage workshop to construct the world's first 20-channel automated peptide synthesis apparatus . This DIY invention enables him to produce "millions of compounds" via combinatorial chemistry and positions his lab at the forefront of a new approach.

Date/Place: 1987 (approx.), in a neighbor's garage workshop in Copenhagen, Denmark (during Meldal's early career at Carlsberg Laboratory).

People: Morten Meldal; (neighbor with a home workshop; possibly one or two colleagues helping). Mentor Klaus Bock had recruited Meldal to Carlsberg Lab around this time to build up biomolecular synthesis capabilities .

Concrete details: Using a solid Teflon block, Meldal personally drills 20 small reaction columns into it . He assembles the machine by hand with basic tools, creating a device that can synthesize 20 different peptides in parallel. This crude-looking but effective prototype, built "in the garage of my neighbor," became the basis for a published method to create massive "split-mix" peptide libraries . He later improved the design so that *"in one week, [we] can synthesize two million compounds"* on an updated machine . The garage workspace, smell of Teflon shavings, and whir of a drill underscore the improvised nature of the breakthrough instrument.

Why it matters: This scene exemplifies Meldal's resourcefulness and commitment to innovation. The multi-column synthesizer was crucial for combinatorial chemistry – it allowed his lab to quickly generate and test vast libraries of molecules . That capability directly set the stage for his search for "clean" reactions; the combinatorial experiments needed perfectly efficient chemistry so that millions of compounds could be made without purification . In Meldal's words, *"this is what led up to the click chemistry"* – the garage-built machine enabled the work that prompted his most famous discovery . Career-wise, this achievement established him as a creative problem-solver and helped secure major grants (e.g. the Danish National SPOCC Center in 1997) to develop quantitative reactions on solid support .

Evidence quality: Primary – Meldal recounted this story in a Nobel interview (object to museum) . Supporting context from his autobiography confirms his mechanical skills and that Klaus Bock enlisted him to strengthen synthesis tech at Carlsberg . The specifics (year ~1987, 20-column block in a garage) come straight from him.

Myth risk: Low – It's a "built in a garage" story, which could sound legendary, but it's documented by Meldal himself with concrete details . The claim that it was first of its kind and enabled millions of compounds is supported by his publications and recollections (not just an apocryphal tale).

Sources: Nobel interview (transcript) ; Meldal Nobel autobiography .

SCENE #3: Accidental Discovery of the 'Click' Reaction

Summary: In the spring of 2001, in a Carlsberg Laboratory fume hood, Meldal and his PhD student Christian W. Tornøe struggle with a stubborn experiment – trying to couple an azido acid chloride to an alkyne attached on a resin. Every attempt "failed" to give the expected product, yielding an odd byproduct instead . Upon careful analysis, they realize the "failure" is actually a stunning success: the copper catalyst has caused an azide-alkyne *cycloaddition*, forming a 1,2,3-triazole ring in quantitative yield at room temperature . Crucially, the reaction is extremely

clean (no side products) and perfectly regioselective (forming only the 1,4-substituted triazole) . In that eureka moment, Meldal recognizes this is the kind of **"quantitative chemical transformation"** needed for combinatorial chemistry – the Holy Grail reaction they had been searching for . They immediately pivot the research focus to this discovery, later termed the Cu-catalyzed azide–alkyne cycloaddition (CuAAC) – the first *"click chemistry"* reaction.

Date/Place: 2001 in the lab at Carlsberg Laboratory (Copenhagen); first publicly presented in June 2001 at the American Peptide Symposium in San Diego . (Published the following year).

People: Morten Meldal; Christian W. Tornøe (PhD student); Caspar B. Christensen (student, collaborator); colleagues in Meldal's group; scientists at the Peptide Symposium (audience for the announcement). Barry Sharpless's group at Scripps was independently discovering the same reaction around this time (though not present, they are parallel actors in this scientific moment).

Concrete details: Tornøe had been attempting a multi-step synthesis on polymer beads – adding a copper(I) salt to promote a difficult acyl-alkyne coupling . Instead of the desired keto-alkyne, he "consistently got a side-reaction" . The mysterious product turned out to be a triazole ring formed by the azide and alkyne clicking together . Meldal recalls *"we fought a brave battle to make this reaction work"* until the surprising result was identified . The highly reactive acid chloride they started with was still intact, and the azide and alkyne had snapped together cleanly . Realizing *"this…reaction had extraordinary properties"* and was *"just what we had been looking for"*, Meldal had Tornøe and Christensen rush to share the news . In San Diego, June 2001, they present a poster or talk showing the "crown jewel" click reaction . One can picture the mix of frustration turning into excitement in the lab, and later the buzz at the conference when a simple copper-catalyzed reaction is shown to reliably "snap" molecules together.

Why it matters: This is the pivotal discovery of Meldal's career. Scientifically, the CuAAC reaction revolutionized chemical biology and drug development by providing a *reliable, easy way to join molecules*, akin to molecular Lego pieces . It has enormous impact: from medicinal chemistry to materials science, it enabled "click chemistry" as a field . For Meldal's trajectory, this discovery (simultaneous with Sharpless's team) earned him lasting recognition and ultimately the Nobel Prize. Narratively, it's the "eureka" turning point – a moment where an apparent failure reveals a new path. It also illustrates his philosophy that important findings often arise unexpectedly when "following the most promising trails" in the lab .

Evidence quality: Primary and strong – Meldal described the discovery in detail in his Nobel lecture and autobiography, naming the students and the symposium . The timing (June 2001) and technical specifics are documented in conference proceedings and the 2002 publications (Tornøe et al., 2002) . We have a first-person narrative of the lab realization and its significance .

Myth risk: Low – This "accidental discovery" story is well-substantiated by Meldal's own accounts and contemporaneous reports. It's not a science lore exaggeration; both Meldal and Sharpless have acknowledged the serendipity. One risk would be oversimplifying it as a single

“lightbulb moment” when in reality it followed extensive experimentation – but Meldal’s telling makes clear it was a process of noticing an unexpected result .

Sources: Meldal Nobel lecture (2022) ; Nobel interview ; El País interview .

SCENE #4: Setback at Carlsberg – The Lab That Went Dark

Summary: In 2005, Meldal faces a major career upheaval. The Carlsberg Research Center, where he had worked for nearly two decades, decides to end fundamental chemistry research to refocus on brewery-related science. Virtually overnight, the thriving research unit he helped build – including the SPOCC center and various spin-off projects – is slated for shutdown . Colleagues and fellow professors depart en masse, and expensive instruments are dismantled. Meldal attempts to adapt, even diverting his expertise to develop a new polymer for biodegradable beer bottles to fit the institute’s new priorities . Despite his efforts, funding for basic research dries up and morale plunges. He is left “somewhat despondent” contemplating his next move .

Date/Place: 2005–2006, Carlsberg Laboratory in Copenhagen, Denmark. (The policy shift was decided in 2005; the chemistry facilities were disassembled over the next couple of years .)

People: Morten Meldal; Carlsberg Research Center leadership (making the decision); fellow professors/researchers at Carlsberg (most left for other jobs); Phaedria St. Hilaire (Meldal’s wife and colleague, who helped lead a spin-off and also eventually left for industry).

Concrete details: In a meeting announcement, Carlsberg Brewery executives declare a “strategic focus” change – brewing science will take precedence, and non-brewing projects lose support . We see labs packing up: advanced NMR machines and mass specs that Meldal once had at his fingertips are removed or sit idle . Many of Meldal’s “in-house collaborators” and students resign or are laid off . With his research funds dwindling, Meldal spends his time trying to repurpose his polymer chemistry knowledge for packaging materials. He even engages in a somewhat ironic task: inventing better plastic for beer bottles . It’s a bleak, anticlimactic scene – the halls that celebrated the “click” discovery just a few years earlier are quiet, and Meldal, who had thrived in an interdisciplinary environment, now finds himself nearly alone and uncertain.

Why it matters: This scene is a dramatic setback that tests Meldal’s resilience. It marks the end of an era – the closure of Carlsberg’s fundamental research meant Meldal had to either transform or leave. It directly leads to his next chapter: after this low point, he accepts a professorship at the University of Copenhagen in 2011 , where he rebuilds his research program in a university setting. The scene highlights challenges scientists can face from institutional decisions beyond their control (an “institutional barrier”). Narratively, it provides tension: even after a big discovery, his path wasn’t smooth. It also underlines Meldal’s commitment to science – he tried to adapt rather than give up on research .

Evidence quality: Primary – Meldal describes this transition candidly in his autobiography . The shutdown of Carlsberg’s chemistry research is corroborated by industry history (Carlsberg Laboratory indeed shifted to brewing science around 2005). His personal feelings

(“despondent”) and interim projects (biodegradable bottle material) come straight from his account .

Myth risk: Low – This is not a glorified tale but a documented institutional decision. There’s little incentive to mythologize a setback. If anything, it’s under-reported publicly. We rely on Meldal’s own description for the narrative; no contradictory versions are known. The risk might be in over-dramatizing it; however, the core facts are well-sourced.

Sources: Meldal Nobel autobiography ; Nobel biography (Carlsberg section) .

SCENE #5: Nobel Prize Call and Celebration

Summary: Early on October 5, 2022, Morten Meldal receives a life-changing phone call from Stockholm. Upon answering, he initially suspects a prank by his students – something he’s “tried before” – but he quickly realizes this is real . The Nobel Committee informs him that he has won the Nobel Prize in Chemistry. He remains cool on the phone, but once it sinks in, he faces an agonizing one-hour embargo during which he cannot even tell his wife . When the public announcement finally comes, he opens his office door to an eruption of applause: dozens of colleagues and students have gathered in the hallway, having guessed the news . An impromptu celebration engulfs the department. Within two hours, the department head organizes a reception in the main hall, where about 300 people cheer and celebrate Denmark’s first chemistry Nobel in 25 years . Meldal is lifted by the excitement, and media rush in for a press conference. In the following days, he’s feted as a national “science rock star” , a role he embraces to inspire young students.

Date/Place: Nobel Prize announcement day (5 October 2022), University of Copenhagen, Denmark. (Call received early afternoon local time, after the Nobel Committee’s decision in Stockholm.) Nobel Award ceremonies in December 2022 in Stockholm (subsequent context).

People: Morten Meldal; members of the Nobel Committee (on the call); Meldal’s wife (Phaedria, whom he had to wait to tell); his students and colleagues at University of Copenhagen (gathered outside his office and at the reception); Head of Department; Danish press. Carolyn Bertozzi and Barry Sharpless (co-laureates) are also being announced, though not present with him.

Concrete details: Meldal’s phone rings and three voices from the Nobel Committee are on the line . He feels his “legs and body start shaking” (as he later told one reporter) as they convey the news. He is instructed not to share the news for about an hour until the official announcement . Alone in his office, he impatiently watches the clock. The moment he’s allowed, he steps out to find *“like 50 people in the corridor outside, all cheering”* . Faces of students, lab techs, and professors light up, and they shower him with applause and hugs. Shortly after, he’s front and center at a press conference, camera flashes going off. That same afternoon, a larger celebration fills a hall with hundreds of well-wishers, sipping champagne (as is tradition) under Danish flags. Meldal notes that *“in Denmark, it’s not such a usual thing to get a Nobel Prize…25 years since we last got one”*, so everyone treats it as a national event . He is even compared “to a rock star” in the Danish media – a notion he modestly downplays: *“I wouldn’t say a rock star,*

but a science star?" . In the following weeks, he finds himself suddenly in demand, juggling his research with newfound public engagements: talks to school kids, interviews (like the one this transcript is from), and the Nobel ceremony festivities in Stockholm in December.

Why it matters: This scene provides closure and contrast: the solitary, meticulous experimenter is thrust into the limelight. It humanizes the aftermath of scientific achievement – the disbelief, the joy of peers, and the pride of a nation. For a narrative, it's a celebratory climax, showing how Meldal's decades of work (from the barley field curiosity to garage tinkering to click chemistry) culminate in the highest recognition. It also reveals Meldal's character: he initially suspects a joke (showing humility and perhaps a sense of humor), and when celebrated, he immediately thinks of inspiring "young people…to get into science" rather than basking in fame .

Evidence quality: Primary – Meldal himself describes the call and immediate aftermath in an interview during Nobel week . Multiple news outlets and the University of Copenhagen documented the celebration (e.g., photos of the spontaneous gathering). His quoted reactions ("probably a prank…quickly heard it was serious") are first-hand . All key details (crowd size, timing, 25-year gap for Denmark) are directly from Meldal or official sources .

Myth risk: Low – Nobel calls and celebrations are routine to document and there's little exaggeration needed. Meldal's account is straightforward. A minor detail like thinking it was a prank could become a charming anecdote retold often, but it originates from him . As long as sources are cited, there's no myth-building here.

Sources: Nobel interview transcript ; NobelPrize.org news .

B) **DIRECT QUOTES (3–7)**

Quote 1: *"My philosophy is that you cannot make any new discoveries behind the office desk. You make them in a fume hood."*

Context/date: From an interview during Nobel Week, 6 Dec 2022 – Meldal explaining what he enjoys about hands-on science.

Source (full citation): Nobel Prize Outreach. (2022, Dec 6). *Interview with the 2022 Nobel Prize laureate in chemistry, Morten Meldal* (transcript) . NobelPrize.org. (Emphasizing the importance of laboratory experimentation in making discoveries.)

Quote 2: *"If you have a problem, go for it. Don't think that this is for the workshop, or for a physicist, or for somebody with optics. Go for it yourself."*

Context/date: Same Nobel Week interview (Dec 2022), describing his advice to young researchers – reflecting on how he built his own equipment (the peptide synthesizer) instead of waiting for others.

Source (full citation): Nobel Prize Outreach. (2022). *Interview with Morten Meldal* . NobelPrize.org. (Meldal's credo of being proactive and hands-on in solving research problems.)

Quote 3: *"I'm very fortunate to have an ability to forget about failures. I just go on. I usually have several things going on at the same time and just go with the things that seem to show success."*

Context/date: Nobel Prize interview, 6 Dec 2022 – in response to a question about coping with failure in research.

Source (full citation): Nobel Prize Outreach. (2022). *Interview with Morten Meldal* . NobelPrize.org. (Highlighting Meldal's resilience and experimental persistence.)

Quote 4: *"The only way [science] can excite you is by you knowing about some of the parts that are happening, and you're curious about the things you don't know… It's the curiosity and the engagement in the problem that makes a good scientist."*

Context/date: Nobel interview (Dec 2022), giving advice to students – stressing intrinsic curiosity over ambition for titles.

Source (full citation): Nobel Prize Outreach. (2022). *Interview with Morten Meldal* . NobelPrize.org. (Meldal explaining that passion for the problem itself, not external rewards, drives scientific success.)

Quote 5: *"With click chemistry, we are able to take different functions from nature and put them together like a Lego building set."*

Context/date: Interview with *El País*, June 16, 2023 – Meldal responding to a question about whether his chemistry imitates nature.

Source (full citation): Domínguez, N. (2023, June 16). *Morten Meldal… 'Click chemistry will help us defeat cancer and Alzheimer's'* . *El País*. (Meldal uses a vivid analogy to explain how click chemistry connects molecular pieces easily, akin to LEGO blocks.)

Quote 6: *"Since my PhD studies, I have regularly played a 12-string guitar or a 6-string electric guitar in the evenings, just before bed, as a way of unwinding and preparing my mind for innovative solutions… I am convinced that great scientists may also often be fantastic artists and cooks."*

Context/date: Autobiographical Nobel profile, published 2022/2023 – Meldal reflecting on the role of creative hobbies (music, art) in his life and approach to science.

Source (full citation): Meldal, M. (2023). *Morten Meldal – Biographical* . NobelPrize.org. (In his own written words, Meldal describes how artistic pursuits like playing guitar enhance creative thinking in his scientific work.)

C) **HUMAN REALITY DETAILS (8–15 bullets)**

- **Family background shaped his outlook:** Born in 1954 near Copenhagen, Morten was the eldest of three. His mother was a creative schoolteacher-artist (famously making paintings incorporating things like *"three dead frogs"*), instilling in him a sense of curiosity and creativity . His father was a strict businessman (leader at Philips Denmark) who taught discipline – for example, requiring Morten to account for every krone of his allowance or else lose it the next week . This mix of free-spirited curiosity and structured responsibility influenced Meldal's personality and career approach. — **Source:** Nobel autobiographical sketch

- **Nature was his playground:** As a child, Meldal spent summers on his grandparents' farm on Bornholm island, roaming forests, meadows, and beaches from morning until night . He collected butterflies, rocks, flowers – "everything you could possibly collect" – with his parents during Nordic camping trips . These experiences gave him a lifelong love of nature and a "burning desire to understand" how the natural world works . He recalls lying in barley fields pondering big questions about life and evolution, essentially a young scientist in the making . — **Source:** Nobel interview

- **Rebellious yet self-motivated student:** In high school (Rungsted Gymnasium), Meldal rebelled against his father's strict expectations that he go into business . He admits he "was not too attentive" to studies and got middling grades . To instill discipline, his parents made him take a job at 15 stacking bricks in a factory – he jokes that carrying "12 tons a day" gave him "extraordinarily long arms" as a physical souvenir . This independence phase helped him realize his true passion was in science, not business or finance, even if it meant breaking family tradition . — **Source:** Nobel autobiographical sketch

- **Artist and musician at heart:** Meldal has been deeply involved in creative arts alongside science. He learned guitar as a schoolboy and maintained music as a hobby throughout his life . In the late 1970s, he even lived in a commune of artists, where he met his first wife, Italian ceramist Sandra Davolio . He painted and played music with friends (some of whom, like Henrik Schütze, remained lifelong friends and fellow artists) . To this day he plays in a rock band and *"usually practice on Mondays"*, often on guitars **he built himself** . Meldal believes these artistic pursuits *"train associative thought"* and make him a more creative problem-solver in science . — **Source:** Nobel autobiography ; Nobel interview ; El País interview

- **DIY ethos in work and home:** Meldal is known for being exceptionally hands-on. His father had a home workshop and taught him carpentry and fixing things by example . Morten applied those skills in science – building lab instruments (like his peptide synthesizer) himself – and in his personal life. Notably, after he married Dr. Phaedria St. Hilaire (a fellow scientist) in 2000, they *"stumbled upon a small, old house in a beautiful nature reserve"* near Copenhagen on a bike ride in 2001 and decided to buy it . They spent over a year **renovating it themselves** — the second home he rebuilt with his own hands . Meldal even set up a home chemistry lab in a previous house so he could tinker

with experiments at night and on weekends while being near his family . This do-it-yourself approach permeates his life. — **Source:** Nobel interview ; Nobel autobiography

- **Collaborative and entrepreneurial:** Meldal's lab culture has been described as creative and collaborative. In the late 1990s he co-founded a biotech startup named **Combio** with a friend, to apply his combinatorial peptide technology to drug discovery . Combio provided jobs for many of his students and was successfully sold to a larger company after three years . He later led or inspired other ventures (like **VersaMatrix** for advanced resins, and **2cureX**, a still-active company for oncology screening) under Carlsberg's "Biosector" incubator . This shows his drive to see laboratory discoveries translated into practical applications. — **Source:** Nobel autobiographical sketch

- **Mentorship and family:** Meldal's personal and professional lives intertwined. His second wife, Phaedria, worked in his lab at Carlsberg and was *"instrumental in establishing the technology"* for some of their projects . He credits her as a cherished confidant and "sparring partner" on all matters – scientific and personal . They had a son, Ajani, in 2003 , who appears in photos with a chemistry set at age 4 – hinting at a playful early exposure to science. Meldal's colleagues and students often gathered at his home for social events, blurring the line between work and family in a supportive way . His mentor Klaus Bock is also a recurring figure; Klaus's support in the early career (recruiting him to Carlsberg Lab) was something Meldal later paid forward by fostering young scientists in his own group . — **Source:** Nobel autobiography

- **Lifelong curiosity and optimism:** Those around Meldal note his enthusiasm and positive outlook. He often says "chemistry is everything," conveying an almost childlike wonder at the world . Even after major achievements, he stayed grounded; for instance, upon winning the Nobel, he immediately engaged in outreach, visiting school science clubs to inspire *"fourth and fifth graders"* . He pushes for better visualization and teaching of chemistry to young people, believing that if they can *"see"* the molecular world (through animation or metaphor), it will ignite their imaginations just as nature did for him as a boy . — **Source:** Nobel interview

D) **SCIENCE CONNECTION (Intro Biochem)**

- **Best-fit topic:** Biochemical Methods – specifically, bio-orthogonal labeling and click chemistry in biochemistry.

- **Precise contribution:** Morten Meldal co-discovered the copper-catalyzed azide–alkyne cycloaddition (CuAAC), the quintessential "click chemistry" reaction that **joins two molecules quickly, selectively, and in aqueous conditions** .

- **Bridge explanation (for intro students):** In biochemistry, we often need to attach a probe or join two biomolecules without disrupting the rest of the system. *Click chemistry*

is like a molecular **snap-together tool**: if one molecule has an **azide (-N_3)** group and another has an **alkyne (-C≡C-)** group, a copper(I) catalyst will "click" them together to form a **triazole ring**, linking the molecules . Meldal's CuAAC reaction is highly efficient (near 100% yield) and **bio-compatible**: it can be done in water, at room temperature, and the azide/alkyne groups generally don't react with the functional groups in proteins or DNA . This means scientists can attach things like fluorescent dyes to a specific biomolecule inside a cell by incorporating an azide on the biomolecule and adding an alkyne-dye – the two will find each other and click, without side reactions . Because the reaction is so favorable energetically and produces no byproducts, it's often called "green" and clean chemistry . In short, Meldal's work provides a robust chemical trick to label or assemble biomolecules, which is now used for everything from tracking cell-surface sugars to developing new drugs .

- **Scientist's own framing/analogy:** Meldal likens click chemistry to **Lego** building blocks – each piece clicks together firmly and specifically . He notes that with click reactions, chemists *"take different functions from nature and put them together like a Lego building set,"* making it straightforward for even biologists to pick pieces from a catalog and assemble new molecules . (Barry Sharpless, co-founder of click chemistry, called it like fastening a *"seatbelt buckle"*, emphasizing its ease .)

- **Sources:** Meldal interview (NobelPrize.org, 2022) on click reaction properties ; *El País* interview 2023 (Lego analogy) ; Science News (Nobel explainer) ; PNAS profile of laureates (selectivity of azide-alkyne) .

E) **COMPLICATIONS OR CONTROVERSIES**

- **Item: Independent Discovery of CuAAC (2001)**

  What's agreed: The copper-catalyzed azide–alkyne "click" reaction was discovered independently and almost simultaneously by two parties – Morten Meldal's team and Barry Sharpless's team – in 2001 . Sharpless had published a 2000 manifesto coining "click chemistry" and was searching for a perfect example; Meldal's experimental results and Sharpless's approach converged on the same reaction within months of each other . Both groups share credit, and both Meldal and Sharpless (along with Carolyn Bertozzi for related work) were jointly awarded the Nobel Prize in 2022 .

  What's disputed: There is little dispute between the scientists themselves – it's recognized as a case of simultaneous discovery. If anything, some in the community noted Sharpless's conceptual contribution versus Meldal's serendipitous lab finding. Sharpless introduced the *idea* of click chemistry (comparing it to a seatbelt buckle "click") , while Meldal demonstrated the **first practical click reaction** (CuAAC) with experimental evidence . Both published in 2002. No open conflict or priority fight emerged; they cited each other's work respectfully. The Nobel Committee explicitly honored both, diffusing any credit controversy .

Sources: *El País* interview (2023) – background on Sharpless's 2000 concept and simultaneous discovery ; Nobel Prize press materials .

- **Item: Carlsberg Laboratory Closure (2005)**

  What's agreed: In 2005 Carlsberg Brewery's management decided to refocus the Carlsberg Research Center on brewery-related research, terminating its funding for basic chemistry and biology. This led to the shutdown of projects like Meldal's combinatorial chemistry program and the disbanding of his institute at Carlsberg . Most researchers, including many of Meldal's colleagues, left to find positions elsewhere . Meldal tried to stay by aligning with applied goals (e.g., materials for beer packaging), but by 2011 he moved to academia.

  What's disputed: There wasn't a public controversy, but it was an internal conflict between scientific agenda and corporate priorities. Some might debate whether Carlsberg's decision was short-sighted (losing a world-class chemist) or necessary for their business. Meldal diplomatically notes the challenge but doesn't publicly criticize Carlsberg's leadership in his memoir – he pivoted to continue his work at University of Copenhagen . Essentially, the "dispute" was science versus industry focus, with Meldal on the side of fundamental research. This episode is not widely reported outside of his own account, so it's not a scandal, just a notable hurdle.

  Sources: Meldal's Nobel autobiographical chapter on Carlsberg's policy change ; personal narrative of adapting research goals .

F) **"NOT ON WIKIPEDIA" FINDS (5+)**

- **Mother's frog art** — Morten's mother, Hanne, was an artist who once made a painting using three **dead frogs** as part of the artwork . This bizarre and memorable detail exemplifies the creative, non-conventional environment in which he grew up (and is not mentioned on his Wikipedia page). — *Source: Nobel Prize biographical article*

- **Allowance ledgers** — As a child, Meldal had to keep **meticulous accounts** of his weekly allowance spending to satisfy his father . If he failed to itemize the expenses, he wouldn't get next week's allowance. This strict exercise in bookkeeping is a distinctive family anecdote illustrating the discipline ingrained in him (not found on Wikipedia). — *Source: Nobel Prize biographical article*

- **Home lab for work-life balance** — In the 1980s, Meldal and his first wife converted part of their home into a **mini chemistry lab** so he could "test out ideas and do experiments" at home in the evenings, rather than staying late at the institute . He literally brought his research home to spend time with family. This humanizing detail of a home laboratory is

not on Wikipedia. — *Source: Meldal autobiography on NobelPrize.org*

- **Samba road trip** — Around the time of the click chemistry breakthrough, Meldal took a trip to **Brazil with a samba band**, traveling by bus, playing percussion in local communities for food and lodging . He describes it as a "big experience" that combined adventure with music. Such a colorful sidelight – a future Nobelist busking in Brazil – doesn't appear on his Wikipedia page. — *Source: Nobel Prize interview transcript*

- **Builds his own guitars** — Meldal is an avid musician who not only plays in a **rock band** but also **builds his own guitars** to play on stage . This DIY hobby parallels his hands-on approach in the lab. It's a distinctive personal fact (mentioned in a 2023 interview) that isn't captured in the Wikipedia profile, which sticks to his scientific achievements. — *Source: El País interview*

- **Youthful Marxist and expulsion from home** — In his university years, Meldal briefly joined a **Marxist student group**, leading to heated political debates at home. It got to the point where his parents politely asked him to find other housing to avoid influencing his younger siblings . This anecdote about his rebellious intellectual phase and family dynamics is not on Wikipedia, yet it sheds light on his independent thinking and character development. — *Source: Nobel autobiographical sketch*

G) **SOURCES CONSULTED (Full citations)**

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5. **Science News – Nobel Prize 2022 Explainer** – Rosen, M., & Ogasa, N. (2022, Oct 5). *A way to snap molecules together like Lego wins 2022 chemistry Nobel.* Science News.

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6. **Kiessling PNAS Profile (2023)** – Kiessling, L. L. (2023). Profile of Carolyn R. Bertozzi, Morten Meldal, and K. Barry Sharpless: 2022 Nobel laureates in Chemistry. *Proceedings of the National Academy of Sciences, 120*(35), e2308367120. https://doi.org/10.1073/pnas.2308367120 (Contains background on click chemistry development and each laureate's perspective)

7. **European Science-Media Hub Interview (2023)** – European Parliament ESMH. (2023, Dec 20). *Prof. Morten Meldal: "Science should be done without constraint."* [Interview]. Retrieved from https://sciencemediahub.eu/2023/12/20/prof-morten-meldal-science-should-be-done-without-constraint/ (Used for Meldal's views on research freedom and serendipity in science.)

Additional primary references: Tornøe, C. W., Christensen, C., & Meldal, M. (2002). *Peptidotriazoles: Copper(I)-catalyzed 1,3-dipolar cycloadditions on solid-phase*, presented at the American Peptide Symposium (2001) ; and Sharpless et al. (2002) on click chemistry concept (for historical context).

H) **FLAGGED "MYTH-RISK" DETAILS**

(No major "too perfect" anecdotes were identified in the sources above. The dramatic events (e.g., the accidental discovery of CuAAC, or the Nobel call surprise) are well documented by Meldal himself rather than second-hand lore. All precise timestamps and quotes have been backed by primary sources. One should still take care not to romanticize the discovery as pure luck without the context of the careful analysis Meldal's team did – the narrative should reflect that the "Eureka" moment was born out of meticulous experimentation, not just a random whim.) If any popular retellings omit Christian Tornøe's role or Sharpless's simultaneous discovery, that would be a slight distortion; however, in our evidence, those facts are clearly included . Overall, no widely repeated anecdote was found to be unsupported by the sources at hand, so no specific high-risk myth needs debunking here.

LENS RECOMMENDATION

- **Eureka/Discovery: 3** – The story of discovering the click reaction is richly documented with a clear "aha" moment (the unexpected triazole product) and concrete lab details . This can be the dramatic centerpiece of a narrative.

- **Failure/Setback: 3** – The 2005 closure of Meldal's lab (and even the initial "failed" experiment that led to success) provides a well-defined setback that changed his trajectory . It's pivotal and well-sourced, showing how he persevered and moved to a

new environment.

- **Mentor/Relationship: 2** – Klaus Bock's mentorship and recruiting Meldal to Carlsberg is noted , and his partnership with his wife Phaedria is meaningful , but specific single turning-point moments from these relationships are less vividly documented. (Mentorship is present, just not one standout scene beyond general support.)

- **Controversy/Conflict: 0** – No substantial disputes or conflicts in the narrative. Meldal's story is largely collaborative and positive. Independent discovery with Sharpless was amicable , and any institutional conflict (Carlsberg) was one-sided. There's no personal controversy to center a narrative on.

- **Human Detail: 3** – The evidence includes many vivid personal details (music, art, DIY projects, childhood explorations) that illuminate Meldal's character . These can be woven into the story to give it depth and are strongly sourced.

- **Last Experiment/End of career: 3** – A late-career scene, the Nobel Prize call and celebration, is very well documented and provides a satisfying closure with emotional resonance . Meldal is still active, but this serves as a "culmination" scene that would work well at the end of a profile.

Primary recommended lens: Eureka/Discovery

Why it fits: Meldal's profile can be driven by the excitement of scientific discovery. The CuAAC "click" reaction episode (Scene #3) is highly compelling and central to his significance . We have strong material to depict the eureka moment and its buildup (the need for a perfect reaction, the frustration, then the breakthrough). Other scenes support this lens by providing context: Scene #2 (building the synthesizer) sets up why he needed the discovery, and Scene #4 (lab shutdown) underscores the value of that discovery in hindsight. The narrative can follow how curiosity and hands-on experimentation led to a serendipitous breakthrough.

Best opening scene: Scene #3, the accidental discovery in the lab, would grab attention if opened in medias res (e.g., Tornøe staring at an NMR readout of the "wrong" product). Alternatively, one could open with a bit of Scene #1 (child Morten's sense of wonder) to establish the curiosity theme, then jump to Scene #3. Scene #3 is the anchor.

One quote to use early: Quote 1, *"You cannot make any new discoveries behind the office desk… You make them in a fume hood."* – This can set the tone that real discovery for Meldal happens through experimental work, foreshadowing the eureka moment in the lab.

Cautions: Ensure the narrative doesn't oversimplify the science – clarify that Meldal recognized the significance of the "failed" reaction through careful analysis, not sheer luck alone. Also, note that Sharpless independently made the same discovery to avoid implying Meldal was solely responsible (though in a narrative you can focus on his personal experience). There's no real

myth to bust, but avoid the trap of a single “flash of genius” trope; emphasize his persistent *process* that led to the breakthrough .

Backup lens: Human-Creative Side (Personal Journey)

If not focusing solely on the discovery, an alternative lens is to highlight Meldal’s **creative, hands-on personality** and how it shaped his science. This would score nearly as high (Human Detail = 3, Eureka = 3, Failure = 3) and can be within one point of the Eureka lens. The narrative would thread together scenes of him building things (garage synthesizer in Scene #2), playing music (perhaps mentioning the samba band anecdote), and nurturing curiosity (Scene #1’s barley field and Scene #5’s outreach after Nobel). His science is then an extension of this inventive spirit.

Why it fits: We have abundant vivid evidence of Meldal’s unconventional path – artist mother, building gadgets, music improvisation aiding his thinking . These details can engage readers and inspire students by showing the human side of a scientist. Scenes #1, #2, and #5 especially support this lens, showing his journey from a curious child to a resourceful inventor to a celebrated scientist who still sees himself as an ordinary person with hobbies.

Best opening scene: Scene #2 (1987 garage invention) could be a fun cold-open – imagine him drilling a Teflon block while others order expensive equipment, immediately illustrating his DIY ethos. Or Scene #1, with young Morten in nature, to set a reflective tone.

One quote to use early: Quote 6, *“I play the guitar… as a way of preparing my mind for innovative solutions… great scientists may also often be fantastic artists.”* . This directly ties his human hobby to his scientific creativity and would signal to readers that the story is as much about the person as the discovery.

Cautions: The risk here is not to let the personal anecdotes overshadow the explanation of why his scientific work matters. We’d need to still clearly connect the dots to the biochemistry (perhaps by using the DIY synthesizer leading to click, etc.). Additionally, ensure each personal detail used is clearly sourced (to avoid seeming like fluff) – we have done that. This lens should complement, not trivialize, the science.

Final Submission:

A glass fume hood hums at the Carlsberg Laboratory in Copenhagen. **Morten Meldal** leans in while his PhD student, Christian W. Tornøe, keeps pushing the same stubborn chemistry forward on polymer beads, hoping an azido acid chloride will couple cleanly to an alkyne on a resin. Instead, every attempt "failed," producing the same unexpected byproduct.

Meldal's group has been chasing a reaction that behaves like a machine part, a transformation so clean and predictable that you can repeat it thousands or millions of times without stopping to purify anything. Their broader work depends on scale and speed, but the experiment on the bench keeps refusing to cooperate.

Then the "wrong" result starts to look like the right one. The copper they added to help one bond form has quietly driven a different transformation altogether: an azide–alkyne cycloaddition that snaps the two groups together into a 1,2,3-triazole ring, in near-quantitative yield, at room temperature, with no messy side products, and in only one orientation. Meldal remembers the struggle—"we fought a brave battle to make this reaction work"—but the battle ends not with a compromise, but with a new rule. The acid chloride they began with is still intact; the azide and alkyne have simply clicked together, cleanly, as if they had been waiting for permission.

This is the kind of discovery Meldal trusts: earned with hands, not announced from a chair. "My philosophy is that you cannot make any new discoveries behind the office desk. You make them in a fume hood," he says, describing the physical place where surprises have a chance to show their faces. The group pivots immediately, treating the unexpected triazole as the transformation they had been searching for, and they rush to present it publicly that summer.

On paper, the chemistry is simple to say and hard to overstate. If one molecule carries an azide group and another carries an alkyne, a copper(I) catalyst can join them into a triazole link in water, at room temperature, without those groups interfering with the usual functional groups found in biomolecules. That selectivity is the point: it lets researchers attach tags or connect molecular parts without the rest of a complex mixture erupting into side reactions.

Years earlier, Meldal drilled a solid Teflon block into a 20-channel peptide synthesizer in a neighbor's garage—an improvised machine meant to make enormous libraries of molecules quickly. The logic is the same as at the fume hood: if you need a capability, build it; if you hit a wall, keep experimenting until the wall becomes a door. And even later, when corporate decisions at Carlsberg began dismantling fundamental research, Meldal tried to adapt by redirecting his chemistry toward biodegradable beer-bottle materials—what remains constant is the stubborn appetite for making something that works.

Back at the hood, the air system still drones, indifferent. But a small ring of atoms—born from a "failed" experiment that someone bothered to understand—changes what "reliable" can mean in molecular work, and it happens the way Meldal insists it must.